

SOFTWARE REQUIREMENTS SPECIFICATION (SRS)

Unified Cyber Defense Platform — SIEM Command Center

Enterprise Security Information & Event Management System with Real-Time Log Correlation, PCAP Telemetry Parsing, Active Directory Auditing, and Automated SOAR Incident Containment.

Document Title	Software Requirements Specification (SRS)
System Identifier	SIEM-COMMAND-CENTER-v1.0
Standard Baseline	IEEE Std 830-1998 (Recommended Practice for SRS)
Compliance Framework	CompTIA Security+ SY0-701 & NIST SP 800-53 Rev. 5 Controls
Live Console URL	https://unified-cyber-defence.netlify.app/
Source Code Repository	https://github.com/kharedhruva-tech/Unified-Cyber-Defense-Platform-SIEM-Command-Center
Project Author	Dhruva Khare (Lead Security Architect)
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Executive Note: This document establishes the formal software requirements baseline for the Unified Cyber Defense Platform. It outlines system boundaries, multi-tier microservice interactions, API data flow specifications, functional feature criteria, non-functional security constraints, and complete traceability against CompTIA Security+ SY0-701 certification objectives.

1. INTRODUCTION

1.1 Purpose

The purpose of this Software Requirements Specification (SRS) document is to define the complete functional and non-functional requirements for the **Unified Cyber Defense Platform — SIEM Command Center**. This system operates as a centralized Security Information and Event Management (SIEM) and Security Orchestration, Automation, and Response (SOAR) platform designed for enterprise threat monitoring, incident response automation, network traffic analysis, and identity auditing.

1.2 Document Conventions

This document follows the structural standards specified in IEEE Std 830-1998. Requirements are uniquely labeled using the convention **REQ-[MODULE]-[ID]**. Priority levels are defined as **High (Mandatory)**, **Medium (Desirable)**, and **Low (Optional)**.

1.3 Intended Audience & Reading Suggestions

This document is intended for Security Operations Center (SOC) analysts, security architects, software developers, CompTIA Security+ auditors, and system administrators. Readers interested in high-level architectural flow should focus on Section 2, while implementation engineering teams should review Sections 3, 4, and 5.

1.4 Project Scope

The Unified Cyber Defense Platform provides end-to-end security visibility by ingesting normalized logs across heterogeneous environments (Linux hosts, Windows Active Directory Domain Controllers, perimeter firewalls, and network PCAP captures). Key operational capabilities include:

- **Real-Time Event Correlation:** Automated SIGMA pattern-matching engine detecting brute-force spikes, privilege escalation, and lateral movement.
- **Deep Packet Inspection (DPI):** Wireshark-compatible PCAP analysis identifying rogue protocols, DNS tunneling entropy anomalies, and ARP spoofing.
- **Active Directory Security Governance:** Audit engine evaluating Event ID 4625 failures, Kerberos ticket hygiene, and GPO compliance.
- **Interactive 3D & GIS Visualizations:** Three.js WebGL 3D threat globe and Leaflet GIS breach map for geographical IOC tracking.
- **Automated SOAR Dispatcher:** One-click and event-triggered host isolation, firewall IP containment, and Slack/Discord webhook alerts.
- **AI Security Copilot:** Contextual natural language incident breakdown and automated remediation playbooks.
- **Executive PDF & CSV Reporting:** ReportLab PDF engine compiling executive compliance briefings and raw telemetry exports.

1.5 References Baseline

- 1. IEEE Std 830-1998: IEEE Recommended Practice for Software Requirements Specifications.
- 2. CompTIA Security+ SY0-701 Exam Objectives Framework.
- 3. NIST SP 800-53 Rev. 5: Security and Privacy Controls for Information Systems and Organizations.
- 4. MITRE ATT&CK; Framework v14.1 (Enterprise Matrix).
- 5. SIGMA Rule Format Specification for Log Event Detection.

2. OVERALL SYSTEM DESCRIPTION

2.1 Product Perspective & Microservices Architecture

The platform utilizes a decoupled microservices architecture comprising four major operational tiers:

Architecture Tier	Technology Stack	Operational Responsibilities
1. Client Edge & CDN Layer	Netlify Edge CDN, HTTPS TLS 1.3	Global static asset delivery, SSL termination, client caching header control.
2. Presentation Tier	React 18.3, Vite 5, Tailwind CSS, Three.js, Leaflet	Interactive SOC command dashboard, 3D threat globe, real-time log tables, session telemetry cache.
3. Microservices Backend	Python 3.10+, FastAPI, Pydantic v2, SIGMA Engine	RESTful API controllers, log normalization pipeline, PCAP packet parser, AD audit engine, SOAR containment dispatcher.
4. Datastore & Storage	Supabase PostgreSQL, SQLAlchemy ORM, ReportLab PDF	Relational datastore for logs, users, alerts, assets, vulnerabilities, and automated PDF briefing generation.

2.2 User Classes & Role-Based Access Control (RBAC)

The system enforces strict Role-Based Access Control (RBAC) across four distinct operator classes:

User Role Class	Permission Level	System Access & Operational Scope
Administrator (Admin)	Full Administrative Control	User account management, global rule toggling, system configuration, full SOAR containment dispatch.
Security Manager (Manager)	Read/Write Incident Operations	Incident creation and escalation, triage execution, SOAR manual block trigger, report downloads.
SOC Analyst (Analyst)	Read/Write Analytical Access	Real-time telemetry monitoring, PCAP inspection, rule testing, alert investigation.
Security Auditor (Auditor)	Read-Only Governance Access	Read-only access to logs, compliance scorecards, Active Directory audit trends, and executive PDF briefs.

2.3 Operating Environment & Constraints

- **Client Environment:** Modern web browser supporting ES2022 JavaScript, HTML5 Canvas, WebGL 2.0, and CSS Grid (Chrome 100+, Firefox 100+, Edge 100+).
- **Backend Runtime:** Python 3.10+ execution environment hosted on Linux or Docker container infrastructure.
- **Network Constraints:** Requires outgoing HTTPS egress (Port 443) for Supabase DB connections and webhook notifications.

3. SYSTEM FEATURES & FUNCTIONAL REQUIREMENTS

3.1 Authentication & Session Management Module

Description: Provides multi-role authentication issuing cryptographically signed JSON Web Tokens (JWT) using HMAC-SHA256 algorithms and bcrypt password hash verification.

Req ID	Requirement Statement	Priority
REQ-AUTH-01	The system shall authenticate operators using username/email and password credentials via <code>POST /api/v1/auth/login`</code> .	High
REQ-AUTH-02	The system shall issue a signed JWT bearer token upon successful authentication containing role scopes.	High
REQ-AUTH-03	The system shall enforce role-based access restrictions across all API endpoints using custom FastAPI dependencies.	High
REQ-AUTH-04	The system shall implement 15-minute session telemetry caching in <code>sessionStorage`</code> to minimize redundant backend token validations.	Medium

3.2 Log Ingestion & Telemetry Aggregation Engine

Description: Aggregates and normalizes security logs from Linux auth.log, Windows Sysmon/Event Viewer, and perimeter firewalls into standard JSON schemas.

Req ID	Requirement Statement	Priority
REQ-LOG-01	The system shall ingest raw syslog strings and extract structured metadata: <code>timestamp`</code> , <code>log_type`</code> , <code>source_ip`</code> , <code>severity`</code> , and <code>payload`</code> .	High
REQ-LOG-02	The system shall support real-time log pagination and field-specific filtering via <code>GET /api/v1/logs`</code> .	High
REQ-LOG-03	The system shall maintain a high-performance memory stream allowing live telemetry simulation during testing.	Medium

3.3 SIGMA Event Correlation Engine

Description: Evaluates streaming events against active SIGMA detection rules to flag threat patterns (e.g. Brute Force spikes, SSH password sprays, Kerberos ticket anomalies).

Req ID	Requirement Statement	Priority
REQ-SIGMA-01	The correlation engine shall evaluate incoming event rates against configurable threshold windows (e.g. >5 failed logins in 60s).	High
REQ-SIGMA-02	The engine shall assign severity levels (<code>Critical`</code> , <code>High`</code> , <code>Medium`</code> , <code>Low`</code>) and map alerts to MITRE ATT&CK; TTP IDs.	High
REQ-SIGMA-03	The system shall expose rule management endpoints enabling operators to enable, disable, or modify SIGMA rule thresholds.	Medium

3.4 Wireshark PCAP Deep Packet Inspection Service

Description: Inspects network packet streams to detect malicious protocols, high-entropy DNS query tunneling, and ARP spoofing broadcast floods.

Req ID	Requirement Statement	Priority
REQ-PCAP-01	The service shall dissect PCAP stream frames into TCP, UDP, ICMP, and ARP headers via `GET /api/v1/network/pcap`.	High
REQ-PCAP-02	The service shall calculate Shannon Entropy scores on DNS query strings to detect data exfiltration tunneling (>4.2 entropy threshold).	High
REQ-PCAP-03	The service shall detect duplicate IP-to-MAC associations triggering ARP spoofing alerts.	Medium

3.5 Active Directory Security & Governance Auditor

Description: Monitors Windows Active Directory Domain Controller health, tracking Event 4625 login failures, Kerberos ticket hygiene, and GPO compliance.

Req ID	Requirement Statement	Priority
REQ-AD-01	The module shall audit privileged Domain Admin user accounts and flag stale or inactive admin credentials.	High
REQ-AD-02	The module shall track Windows Security Event 4625 (Failed Logon) spikes and calculate user lockout risk scores.	High
REQ-AD-03	The system shall generate compliance scorecards evaluating CIS Benchmark GPO policy enforcement.	Medium

3.6 Automated SOAR Incident Containment Dispatcher

Description: Executes manual and automated threat containment routines, isolating compromised host endpoints and blocking perimeter IP addresses.

Req ID	Requirement Statement	Priority
REQ-SOAR-01	The system shall trigger host isolation routines (`isolate_host`) upon Critical severity alert generation or manual operator invocation via `POST /api/v1/incidents/action/{type}`.	High
REQ-SOAR-02	The system shall dispatch JSON alert webhooks to Slack Block Kit and Discord endpoints upon SOAR action execution.	High
REQ-SOAR-03	The system shall record immutable containment entries in the `AUDIT_LOGS` table for compliance tracking.	High

3.7 Interactive 3D WebGL Threat Globe & GIS Heatmap

Description: Renders real-time threat telemetry geographically using Three.js WebGL spherical mapping and Leaflet GIS breach canvases.

Req ID	Requirement Statement	Priority
REQ-VIS-01	The frontend shall project IP latitude/longitude coordinates onto a 3D rotating globe canvas using Three.js WebGL graphics.	Medium
REQ-VIS-02	The GIS heatmap canvas shall render active breach clusters with ISO country flag resolution via `GET /api/v1/gis/breaches`.	Medium

3.8 Executive PDF & CSV Reporting Module

Description: Dynamic report generation service compiling executive compliance briefings using ReportLab PDF and exporting raw logs to CSV.

Req ID	Requirement Statement	Priority
REQ-REP-01	The system shall dynamically generate executive PDF briefs detailing system posture score, active incident metrics, and vulnerability distributions via `GET /api/v1/reports/download/pdf`.	High

REQ-REP-02	The system shall export raw security event logs to standardized CSV files for offline forensic auditing.	Medium
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4. EXTERNAL INTERFACE REQUIREMENTS

4.1 User Interfaces (UI)

The presentation tier provides a responsive, dark-themed SOC Command Console built with React 18 and CSS design systems. UI components adhere to accessibility standards (WCAG 2.1 AA) and incorporate micro-animations, real-time metrics feeds, dynamic tables, and visual alert indicators.

4.2 Software Interfaces & RESTful API Specifications

The system exposes a standardized RESTful API v1 interface constructed with FastAPI. Below is the API routing contract table:

HTTP Method	API Endpoint Route	Operational Purpose & Return Data
POST	/api/v1/auth/login	Authenticates operator & returns signed JWT bearer token.
GET	/api/v1/reports/summary	Calculates system posture score & SIEM counter metrics.
GET	/api/v1/logs	Fetches normalized log streams with pagination & filter parameters.
GET	/api/v1/threat-detection/alerts	Retrieves active SIGMA-correlated alerts.
POST	/api/v1/incidents/action/{type}	Executes automated SOAR containment (`isolate_host`, `block_ip`).
GET	/api/v1/network/pcap	Returns deep packet inspection dissect data & DNS entropy score.
GET	/api/v1/active-directory/audit	Returns AD Event 4625 statistics & GPO compliance scorecard.
GET	/api/v1/gis/breaches	Returns geolocation breach feed for WebGL & GIS heatmaps.
GET	/api/v1/reports/download/pdf	Compiles dynamic executive security briefing PDF.
POST	/api/v1/notifications/webhook	Dispatches alert payloads to Slack & Discord gateways.

4.3 Communication Interfaces

- **Protocol:** All client-to-server and server-to-database communications MUST use HTTPS with TLS 1.3 encryption.
- **Data Format:** All API request payloads and responses MUST be formatted in strict JSON (JavaScript Object Notation).
- **Webhooks:** Outbound notification webhooks transmit HTTPS POST JSON payloads formatted according to Slack Block Kit and Discord Embed standards.

5. NON-FUNCTIONAL REQUIREMENTS (NFRs)

NFR Category	Requirement Description & Metric Target	Validation Method
Performance	API response latency MUST remain <100ms for standard telemetry queries. Session caching MUST reduce database roundtrips by >80%.	Automated Locust Load Tests

Security & Zero Trust	Passwords MUST be hashed using bcrypt. All API routes MUST validate JWT signatures. CORS policy MUST restrict unauthorized origins.	OWASP ZAP Penetration Audit
Availability	The SIEM platform target uptime is 99.9%. Client dashboard MUST implement React Error Boundaries to prevent total UI crashes during network failures.	Uptime Kuma Health Check
Maintainability	Backend services MUST be structured in modular FastAPI packages. Code coverage MUST exceed 80% across core engines.	Pytest & Coverage Reports
Portability	The entire system MUST be deployable via Docker and Docker Compose across Linux, macOS, and Windows environments.	Docker Multi-OS Verification

6. APPENDICES & CERTIFICATION TRACEABILITY MATRIX

6.1 CompTIA Security+ SY0-701 Domain Traceability Matrix

The table below maps the functional requirements of the Unified Cyber Defense Platform directly to the 5 core domain specifications of the CompTIA Security+ SY0-701 certification framework:

Security+ SY0-701 Domain	Domain Weight	Mapped System Engine & SRS Requirements
1.0 General Security Concepts	12%	JWT Auth Engine & RBAC Enforcement (`REQ-AUTH-01` to `03`), CIA Triad Controls.
2.0 Threats, Vulnerabilities & Mitigations	22%	SIGMA Threat Detection Engine (`REQ-SIGMA-01` to `03`), Vulnerability Scoring Module.
3.0 Security Architecture	18%	Decoupled Microservices Topology, Active Directory Audit (`REQ-AD-01` to `03`), DMZ Microsegmentation.
4.0 Security Operations	28%	Wireshark PCAP Dissector (`REQ-PCAP-01` to `03`), SOAR Containment (`REQ-SOAR-01` to `03`), 3D WebGL Globe.
5.0 Security Program Management	20%	ReportLab PDF Briefing Engine (`REQ-REP-01`), Immutable Audit Logging, CIS Compliance Scorecards.

6.2 Acronyms & Glossary

- **API:** Application Programming Interface — A set of defined rules enabling application microservices to communicate.
- **Bcrypt:** A password-hashing function based on the Blowfish cipher incorporating salt to protect against rainbow table attacks.
- **CVE:** Common Vulnerabilities and Exposures — A dictionary of publicly known cybersecurity vulnerabilities.
- **CVSS:** Common Vulnerability Scoring System — A standardized numerical framework for assessing vulnerability severity.
- **DPI:** Deep Packet Inspection — Advanced network packet filtering examining data payloads at Layer 7.
- **GPO:** Group Policy Object — A collection of administrative settings in Microsoft Active Directory controlling user/computer environments.
- **JWT:** JSON Web Token — An open standard (RFC 7519) defining a compact, self-contained method for securely transmitting information as a JSON object.
- **PCAP:** Packet Capture — An application programming interface and file format for capturing network packet data.
- **RBAC:** Role-Based Access Control — An approach to restricting system access to authorized users based on defined job roles.
- **SIEM:** Security Information and Event Management — A security solution aggregating and correlating log data across IT infrastructure.
- **SIGMA:** Generic signature format for log events, allowing security engineers to describe relevant log events in a standardized format.
- **SOAR:** Security Orchestration, Automation, and Response — Technologies enabling automated incident containment workflows.

DOCUMENT APPROVAL & SIGN-OFF	
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